Jim Gammell

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Research summary PhD researcher with a broad background spanning machine learning, hardware, and applied physics. Doctoral research has focused on deep learning and interpretability for hardware security. I enjoy solving real-world problems using first-principles thinking and theoretically motivated ML techniques. My broader research interests include explainable AI, security, LLMs and agentic systems, and embodied AI.

Education

Purdue University, West Lafayette, IN — Aug 2021 – Aug 2027 (anticipated)
PhD, Electrical and Computer Engineering — Current GPA: 3.97/4.00
Fellowships: ECE Star Recruit Doctoral Fellowship (Aug 2024 – present),
Center for Secure Microelectronics Ecosystem Traineeship (Aug 2021 – July 2024)

University of Colorado at Boulder, Boulder, CO — Aug 2017 – May 2021
BS, Electrical and Computer Engineering; BS, Applied Mathematics — GPA: 4.00/4.00 (*summa cum laude*)
Selected honors: Outstanding Undergrad Research Award (link), Chancellor's Recognition Award

Experience

Purdue University, West Lafayette, IN — Aug 2021 – present
PhD student; advised by Kaushik Roy, Institute of Chips and AI

- Developed deep learning attacks on cryptographic hardware and interpretability techniques for determining which vulnerabilities the attacks exploited from power/EM traces ("side-channel leakage localization").
- Formulated leakage localization probabilistically and proposed a novel adversarial learning method. Developed 4 quantitative evaluation metrics and the first systematic benchmark of 12 localization methods across 6 datasets; published in TMLR.
- Extended this work to realistic-scale full-key attacks on uncropped traces. Conducted large-scale empirical study showing that models with better attack performance can yield worse characterizations of hardware vulnerabilities, and explained the phenomenon through a Shapley value-like mathematical derivation.
- Developed a simple transformer pipeline for full-key attacks on uncropped traces, showing that well-engineered standard architectures can outperform more-specialized designs and identifying weaknesses in common evaluation metrics.
- Collaborated on project to efficiently estimate counterfactual memorization by LLMs. Wrote efficient pipeline for training and analyzing lots of GPT-2 models to validate proposed estimator.

Amazon, Seattle, WA — June – Sept 2026
Applied Scientist Intern; Amazon Security (AmSec)

- Classified devices from network telemetry under real-world challenges: throughput/latency constraints, severe (5 orders of magnitude) label imbalance, generalization under nonstationarity and location-dependent distribution shift.
- Characterized cost/performance tradeoff of contrastive representation learning, tabular foundation models, AutoML, supervised learning w/ transformers, MLPs, LightGBM, random forests.
- Found that tree-based methods and AutoML were Pareto-efficient, some devices can be reliably classified while others cannot, and that strong performance in a balanced setting can collapse to near-zero precision under realistic class frequencies.
- Built production-ready pipeline using CDK, SageMaker, Step Functions, CloudWatch, Athena to classify ∼200M network logs/hour. Manuscript in preparation.

Amazon, Seattle, WA — Sept – Dec 2025
Applied Scientist Intern; Regulatory Intelligence, Security, and Compliance (RISC)

- Trained multimodal (text, tabular) Perceiver-based classifiers leveraging Qwen3-Embedding models on messy data with nonstationarity, missing features, and label imbalance.

- Proposed a mechanism for end-to-end hard retrieval-augmented classification via REINFORCE-based gradient estimator.
- Work selected for technical talk (acceptance rate 53/740) at 2026 Amazon Consumer Science Summit (CSS), and published at ICLR 2026 Workshop on Time Series in the Age of Large Models (TSALM) (link).

Sandia National Laboratories, Livermore, CA May – Aug 2024
URA Summer Graduate Fellow; Combustion Research Facility

- Modeled chaotic partial differential equations with diffusion models. Discovered that models trained on individual PDE states can generate plausible entire temporal rollouts without retraining.
- Wrote efficient pipeline using PyTorch, Numba and Ray Tune. Experimentally tested various data augmentation strategies.

German Aerospace Center (DLR), Braunschweig, Germany May – Aug 2022
DAAD RISE Pro Intern; Onboard Systems

- Implemented and evaluated PPO-based automatic hyperparameter tuning tool for satellite image data analysis; found it was not competitive with baseline genetic optimization under available compute budget.

National Institute of Standards and Technology (NIST), Boulder, CO June 2018 – July 2021
Pathways Intern; Faint Photonics Group

- Proposed small-world-style neural architectures to solve vanishing gradient issues in Equilibrium Propagation (a bio-inspired learning algorithm), leading to journal/conference publications.
- Designed/simulated analog neural net hardware, built software/electronics for superconducting electronics experiments, RF rangefinding, scientific instrumentation.

Selected publications

- **Jimmy Gammell**, Anand Raghunathan, Abolfazl Hashemi, and Kaushik Roy. "Learning to Localize Leakage of Cryptographic Sensitive Variables." *Transactions on Machine Learning Research (TMLR)*, 2026. [Paper] [Code]
- **Jimmy Gammell** and Kaushik Roy. "A Simple Transformer Pipeline for Full-Key Side-Channel Attacks on Uncropped Datasets." *Presented at the OPTIMIST workshop at Conference on Cryptographic Hardware and Embedded Systems (CHES)*, 2026. [Paper] [Code]
- **Jimmy Gammell**, Bishal Thapaliya, Yoon Jung, Riyasat Ohib, Bilel Fehri, and Deepayan Chakrabarti. "Learning to Query History: Nonstationary Classification via Learned Retrieval." *ICLR 2026 Workshop on Time Series in the Age of Large Models (TSALM)*, 2026. [Paper]
- **Jimmy Gammell**, Sonia Buckley, Sae Woo Nam, and Adam N. McCaughan. "Layer-Skipping Connections Improve the Effectiveness of Equilibrium Propagation on Layered Networks." *Frontiers in Computational Neuroscience*, 2021. [Paper]

Selected skills

- **Programming:** Python, C (embedded)
- **ML/scientific computing:** PyTorch, PyTorch Lightning, NumPy, Numba, HuggingFace, Ray Tune, Optuna, LightGBM, AutoGluon
- **Systems/cloud:** Linux, Git, AWS (SageMaker, EC2, CDK, Bedrock)
- **Hardware:** Embedded systems, PCB design, electronics instrumentation, RF systems, electronics lab equipment automation